



How Much Do Fans Love Taylor Swift? Day 1



Taylor Swift wants to find out how much fans enjoyed the Eras Tour shows. She will ask fans, "From 1 to 100, where 100 is the most, how much did you enjoy the concert?" The V.I.P. section she wants to survey has 50 seats (5 rows x 10 columns). The stage runs along the northern edge of the venue (where Taylor is pictured). She wants to take a sample of 10 fans.

1. Method #1: SRS

Take a simple random sample (SRS) of 10 fans. Explain the steps.

Label the seats #1-50

- Use random number generator to select 10 numbers. Ignore repeats
- Select people in corresponding seats.



X			X					X	
	X				X	X			
									X
X			X			X			

2. Method #2:

Randomly choose 2 fans from each horizontal row.

Stratified random sample
strata = rows



	X			X					
		X					X		
				X				X	
X					X				
		X	X						

3. Method #3:

Randomly choose 1 fan from each vertical column.

Stratified random sample
strata = columns



	X				X		X		
		X	X						
X						X			X
								X	
				X					

4. Which method do you think will give the best estimate for the average enjoyment? Why?

Method #2. This method ensures representation from each row. Front row will enjoy more than back row.
choose strata variable as the one with largest effect on response



Now, it's time for the actual data. For each of your samples on the previous page, calculate the average enjoyment. Add your average to the dotplots on the board.



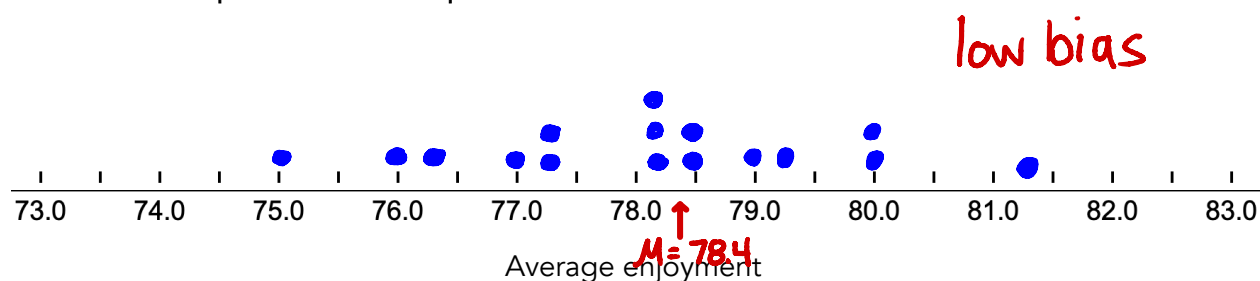
Sample #1: $\frac{92+88+\dots+72}{10} = 82.4 = \bar{x}$

Sample #2: $\frac{89+95+\dots+68}{10} = 79.5 = \bar{x}$

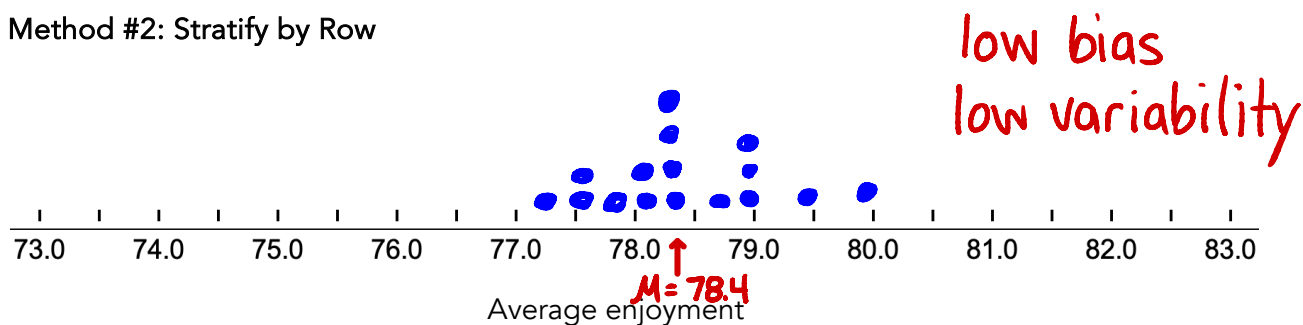
Sample #3: $\frac{89+100+\dots+64}{10} = 81.3 = \bar{x}$

92	89	90	88	95	100	98	93	95	84
82	86	90	88	86	91	90	89	85	83
80	74	80	67	81	82	76	77	74	65
72	68	74	73	70	69	72	70	68	67
69	67	68	68	64	66	63	63	70	68

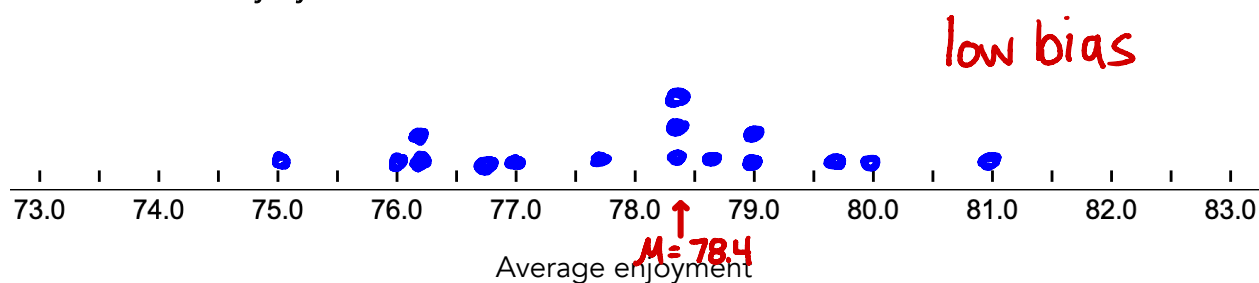
Method #1: Simple Random Sample (SRS)



Method #2: Stratify by Row



Method #3: Stratify by Column



5. Based on the dotplots above, which method will produce the best estimate for the true population average enjoyment? Explain.

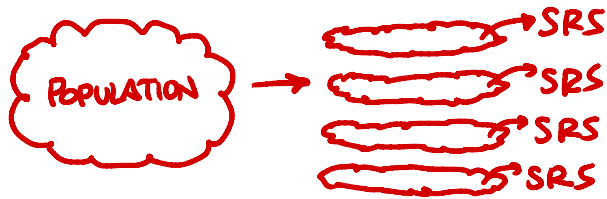
Stratify by row. The dotplot has all estimates close together.

Lesson 2.2 – Stratified Random Samples

QuickNotes

LT1 Stratified random sample

Split the population into groups (strata), then choose an SRS from each strata.



Each strata has individuals with shared attributes or characteristics. (homogeneous grouping)

LT2 Good estimates

A sampling method produces the best estimates if there is

- low bias
- low variability

Check Your Understanding

A principal at a large high school is proposing a new digital hall pass that students would access using their phones. She wants to know what proportion of the student body is in favor of the proposed change. She will take a sample of 100 students.

a. Would this sample be with replacement or without replacement? Explain.

Without replacement. The principal would not want to select the same student twice.

b. Explain how the principal could take a stratified random sample, using grade level as strata.

- Split the student population into groups by grade level.
- For the 9th graders, number them 1 to N.
- Use a random number generator to select 25 numbers (no repeats).
- Select the 9th graders corresponding to these numbers.
- Repeat process for each grade level.

c. Explain why the stratified random sample might produce a better estimate than a simple random sample.

Because students in each grade level might have different opinions about the hall pass. It's possible that most 9th graders are for it, while 12th graders are against it.

stratified would have lower variability